DATA SCIENCE PROJECT

Bike Sharing Demand Prediction

Objective: Predict the hourly number of bike rentals using temporal, calendar, and weather-related features

🎤 **Presentation:** Heike Thöricht

📍 **Bremen,** 23. August 2026

Agenda

The Challenge

Why demand forecasting matters

Our Approach

Data-driven solution

Model Performance

Proven accuracy

Key Insights

What drives demand

Implementation & Next Steps

How to use it

Executive Summary — Ready for Go-Live

Prediction Accuracy

89.7% of demand variance explained — enables confident planning

Forecast Error

±47 bikes on average — manageable margin for logistics

Real-Time Insights

Hour-of-day patterns drive 85% of predictions — actionable for operations

The model is production-ready and will support your planning processes with high confidence.

The Challenge: Why Demand Forecasting Matters

Our Data Foundation

Real-world hourly rental data from the Capital Bikeshare system — a proven operational dataset that mirrors your planning environment.

- **17,379** hourly observations from real bike-sharing operations
- **2 years** of historical data (2011–2012)
- Complete data quality — no missing values

The Business Problem

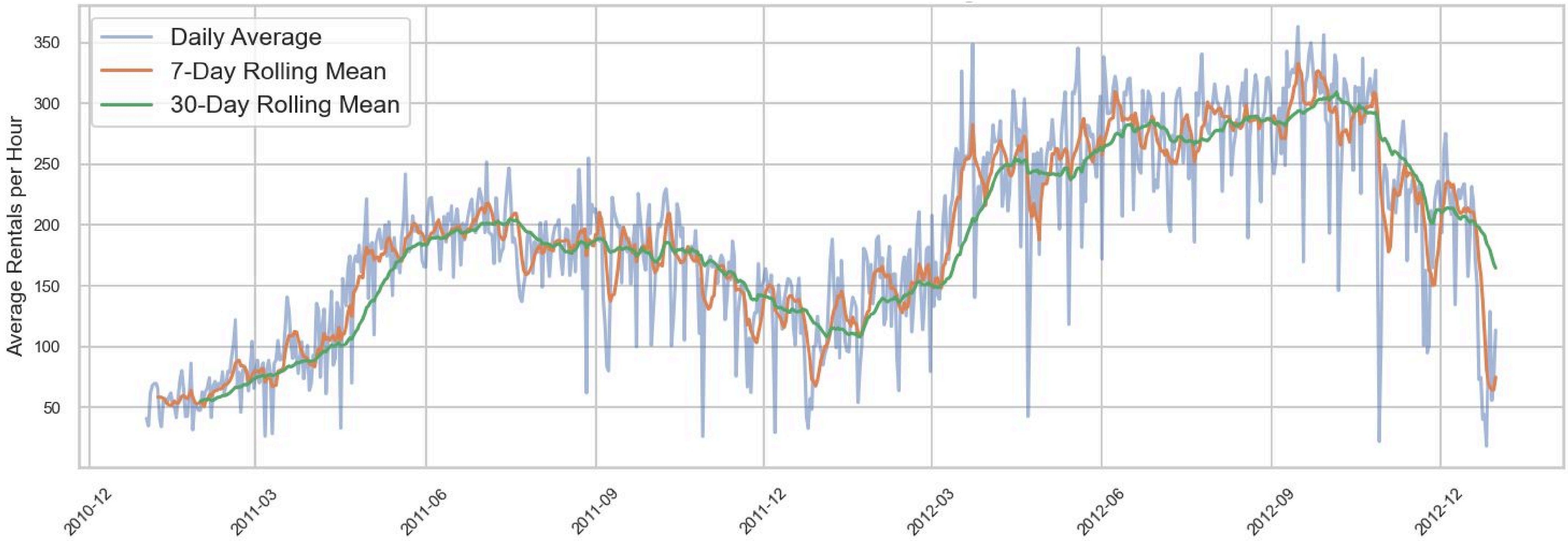
Without reliable demand forecasts, your planning division faces avoidable inefficiencies and operational risk.

- Unpredictable demand creates **logistics and staffing challenges**
- Core question: **How many bikes are needed each hour?**
- **Hourly patterns are the key driver** — not weather or season
- Goal: Accurate hourly forecasts for **smarter resource allocation**

Data source: Fanaee-T, H. (2013). Bike Sharing [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5W894.

EXPLORATORY ANALYSIS

The Rise of Bike Sharing: Emerging Business Opportunities



EXPLORATORY ANALYSIS

Key Insight: Demand Patterns Drive Planning

🕗 Commuter Peaks

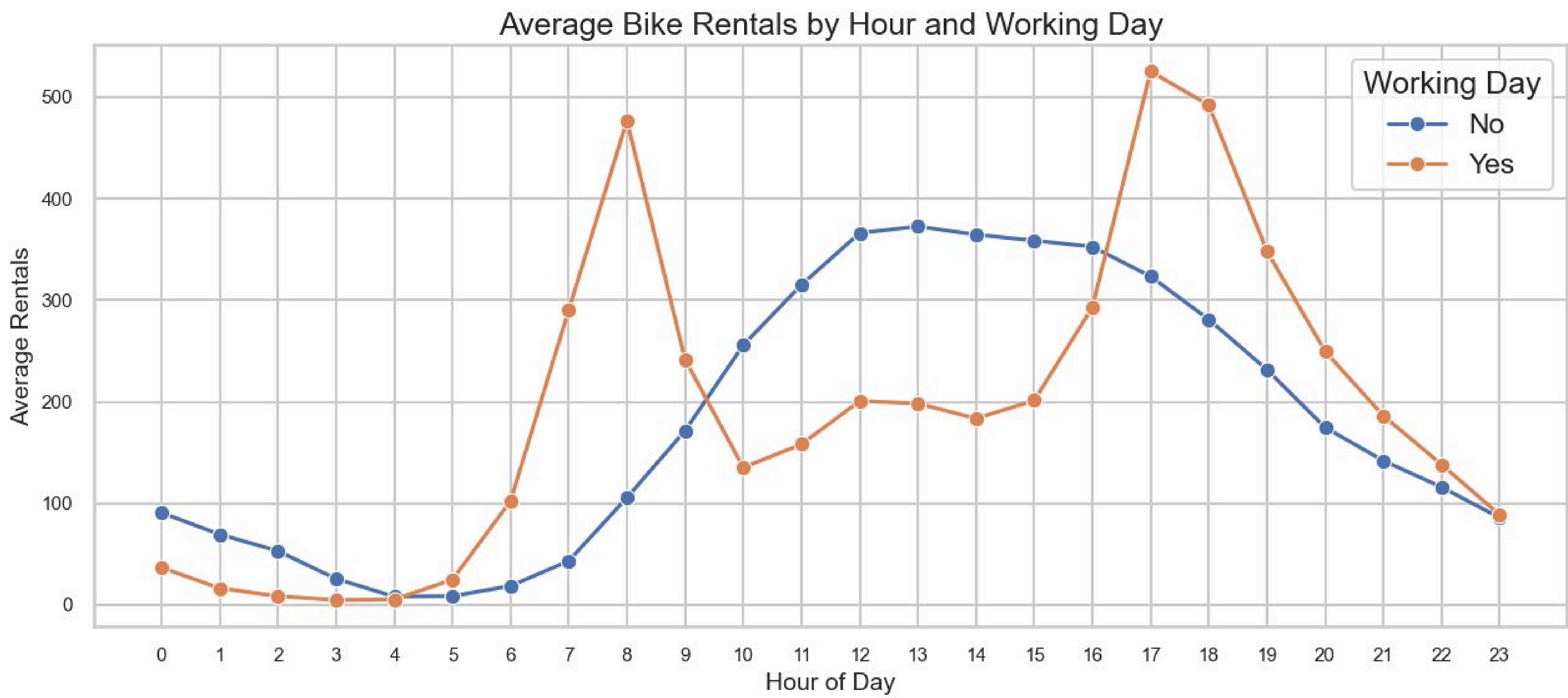
Working days show demand spikes at **8 AM** and **5–6 PM** — align bike availability with commute peaks.

🌤️ Leisure Patterns

Non-working days show midday peaks — different staffing and bike distribution needed

📊 Actionable

Hour-of-day is the **strongest predictor** — enables precise hourly planning



These patterns are consistent and predictable — your planning can rely on them.

MODEL EVALUATION

Model Performance: Proven Accuracy

Comparing our top two models across key performance metrics — XGBoost delivers the best accuracy for production deployment.

Model	Feature Set	R^2	MAE	RMSE
Random Forest	V01	0.880	50.39	76.28
🏆 **XGBoost**	**V01**	**0.897**	**47.35**	**70.68**

🏆 Winner: XGBoost

89.7% accuracy — explains nearly 9 out of 10 demand variations, making it the clear choice for production deployment

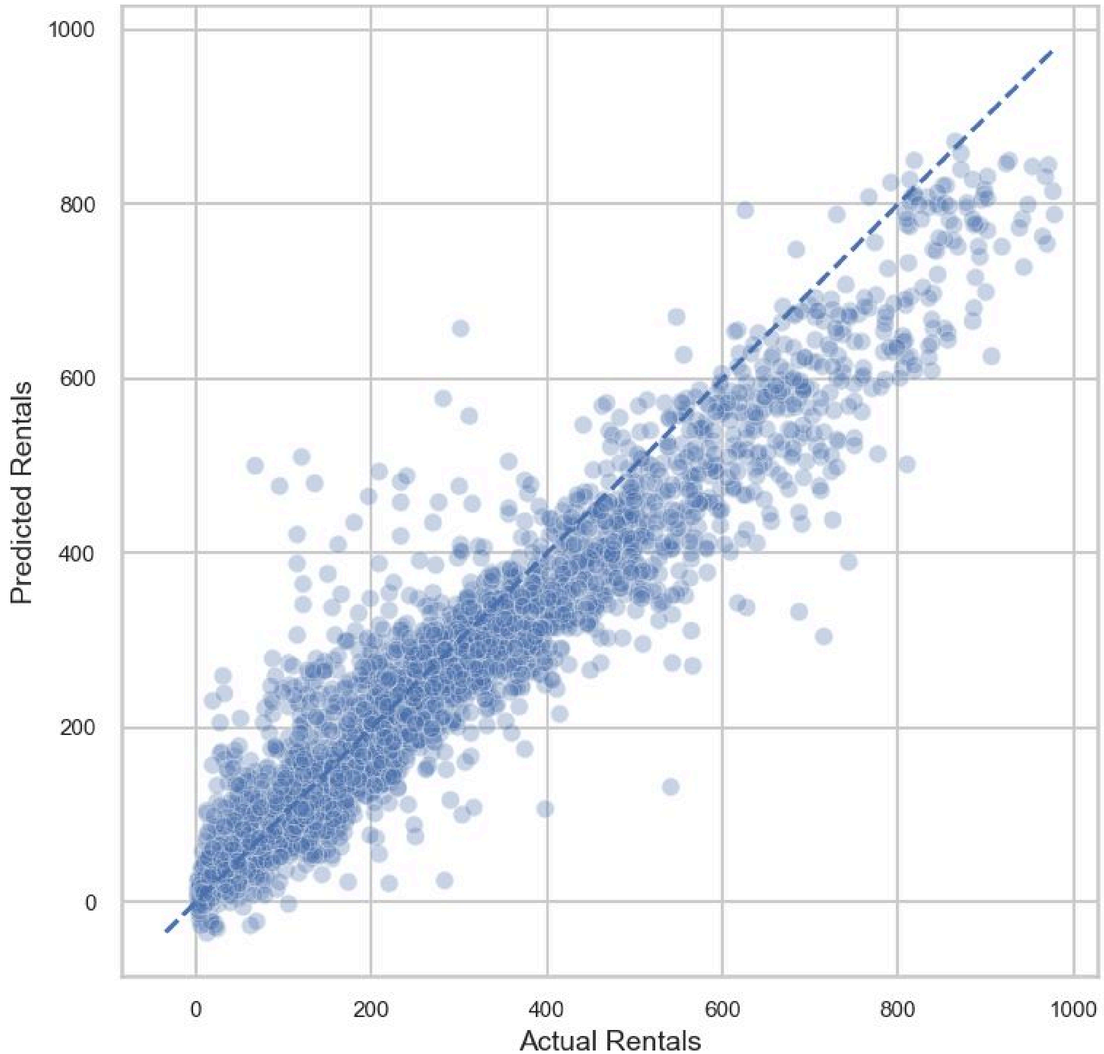
📊 Forecast Error: ±47 Bikes

Average prediction error of just **47 rentals** — an acceptable margin for operational planning and fleet distribution

✅ Production-Ready

Tested on **real time-series data** — proven to generalise beyond training conditions and work in live scenarios

Validation: Model Works in Real Scenarios



✅ Accurate Predictions

Model predictions follow actual demand closely — **reliable for planning**.

⚠️ Peak Hour Challenges

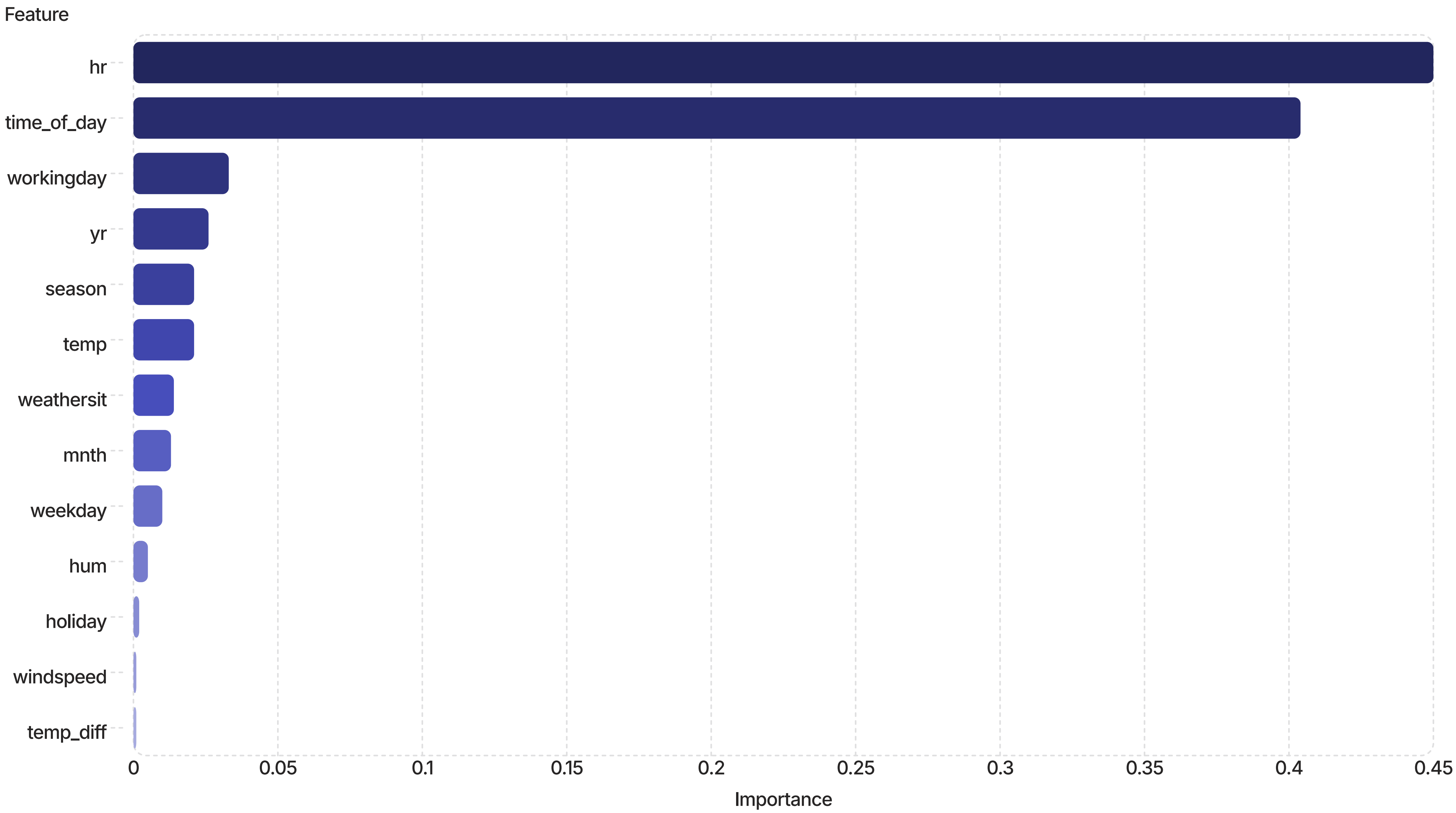
Errors slightly larger during peak hours — expected, but still **within acceptable range**.

🎯 Ready for Operations

Strong overall performance across all demand levels — **suitable for production use**.

The model is validated and ready to support your daily planning operations.

What Drives Demand: The Key Factors



⏰ Time is Everything

Hour-of-day and time-of-day account for **85% of prediction power** — your planning should prioritize hourly forecasts.

📅 Secondary Factors

Weather & Season: Surprisingly minimal impact. Temperature (2.1%) and weather conditions (1.4%) contribute **far less than expected** — hourly patterns dominate.

💡 Actionable

Focus your planning on **hourly patterns**. Weather forecasts are not critical for demand prediction — time-based patterns are what matter.

> The model reveals what actually matters: hourly patterns drive demand. Weather is secondary — **focus on the hour.**

Implementation & Next Steps: How to Use the Model

📊 Forecast Refresh

Update every Monday for weekly planning cycles

Technical approach: Generate daily forecasts (each evening) with hourly predictions for the following day. This provides:

- **Operational usefulness:** Fresh hourly predictions every morning
- **System efficiency:** No need to recalculate every hour
- **Balance:** Captures daily variations without unnecessary overhead

Owner: Data/Analytics team

🔮 Planning Horizon

Reliable 1–2 weeks ahead

Why this horizon?

- **Beyond 2 weeks:** Weather forecasts become unreliable and demand patterns become too uncertain
- **Within 1–2 weeks:** Model captures seasonal patterns and operational variations with confidence
- **Recommendation:** Use rolling weekly forecasts as operational standard; conduct monthly reviews for longer-term trends

Owner: Operations & Planning

🚀 Go-Live Readiness

Production-ready. Needs: API, data pipeline, dashboard

Owner: IT

📈 Success Metrics

Target: MAE < 50 bikes. Retrain quarterly

Owner: Analytics

> **The model is ready. Assign owners, build the pipeline, and start forecasting from day one.**

Questions & Next Steps

Ready to deploy. Let's discuss implementation timeline and resource allocation.

Heike Thöricht

Review forecast dashboard prototype — *this week*

Confirm API requirements with IT — *next week*

Schedule go-live meeting — *within 2 weeks*